Advanced Python Project (Second semester (2025/2026))

Carbon emission dataset (20 Marks) Deadline: 1/6/2026 Instructor Name: Dr. Ali Azwai and Dr. Ala Abuthawabeh

Objective

Students will use **Pandas**, **NumPy**, **Seaborn**, and **Matplotlib** to perform deep data analysis on the **Carbon emission Dataset**.
They will clean, explore, visualize, and interpret real-world Carbon emission data.

Carbon emission Dataset

Dataset: Carbon emission Dataset kaggle (Public)
Link: carbon_emission_dataset_with_Industry.csv

https://www.kaggle.com/datasets/freshersstaff/carbon-emission-forecasting-dataset

Description (Students)

In this assignment, you will work with the **Carbon emission Dataset**, which contains multi-source operational, financial, and ESG-related records from publicly listed companies, designed for carbon footprint accounting and emission prediction. Each entry represents a daily record per company, integrating data from energy usage logs, IoT sensors, supply chain activity, production output, Industry sectors and environmental performance indicators. The target variable is Carbon_Emission_tCO2e, enabling the development and evaluation of forecasting models for emissions management.. You will use Python (Pandas, NumPy, Matplotlib, Seaborn) to perform data exploration, cleaning, visualization, and statistical analysis.

Your task is to complete **all steps below**, answer the required questions, and submit both:

1. A **Jupyter Notebook or A Colad (file)** with code and graphs
2. A **PDF report** summarizing your findings

Grading Breakdown (20 Marks)

Section	Task	Marks
1. Load & Inspect Data	Import dataset, show first rows, info, describe	**2 marks**
2. Data Cleaning	Handle missing values, adjust datatypes	**2 marks**
3. Exploratory Analysis	Tasks: - Compute mean, median, max, and min (Basic Statistics) For	**2 marks**

Section	Task	Marks
	Total_Energy_Consumption_kWh, Renewable_Energy_Consumption_kWh, NonRenewable_Energy_Consumption_kWh, Production_Output_Units, and Supply_Chain_Transport_km -Plot the distribution of each - Create a bar chart for frequency of each - Line plot of Energies over time - Scatter plots: emissions vs temperature emissions vs time	
4. Bivariate & Multivariate **Analysis**	Look at combined relationships. -Compare pairs of emission sources - Correlation heatmap of numerical variables. Box plot of Carbon_Emission_tCO2e_TARGET grouped by Industry_Sectors Pairplot (selected features) Group-based analysis: emissions by region / category (if available)	**2 marks**
5. Time Series Exploration	Convert time column to datetime Plot: emissions over time temperature trend over time Decompose time series: Trend seasonality Rolling statistics: moving average rolling variance	**4 marks**
6. Advanced Questions	Answer 5 analytical questions	**4 marks**

Q1: Which emission sources have highest variability?

Q2: Are emissions normally distributed?
Q3: Which variables are strongly correlated?
Q4: Is there a trend in emissions??
Q5: Do seasonal patterns exist?

7. Submit code on elearning	send Github url	**1 marks**
8. Discussion	Run and presentation (record Video in English)	**3 marks**